Date: 22/10/22

DAA - Assignment 02
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Question #1:

Polygon - intersection:

1- Take two vertices of both polygon in boundary order.

2- Check every edge of polygon P against every edge of polygon Q.

3- for each pair of edges, check wether the two line segments intersect. Use an Orientation to test to identify crossing, touching & collinear cases.

4- If any pair of edges intersects or touches return intersect.

5- If no edges, test whether one vertex of P lies inside Q if yes return "Intersect".
Otherwise return "Do not Intersect"

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2) Pseudocode:

```
Polygon_Intersection(P, Q)
    for each ei in P:
        for each e2 in Q:
            if Sgment_intersect(e1, e2)
                return 'Intersect'.
    if point in polygon(vertex(P), Q):
        return "Intersect"
    if point_in_polygon(vertex(Q), P).
        return "Intersect".

    return "Do not intersect"
```

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1) Question 2:-

Algorithm:-

1- Extract all vertices of star-shaped polygon.
2- Treat these vertices as a set of points.
3 Apply a standard convex hull algorithm, such as Graham Scan, Monotone-chain or Quick-hull.
4- Return the vertices of the resulting convex-hull in boundary order.

2) Pseudocode:-

StarAlgo-convexhull (P):
V: all-vertices of polygon P.
H: Convex-hull (V).
return H:

3) Justification:-

A Convex-hull is smallest convex-set containing all the given points. Since a polygon lies within the Convex-hull of its vertices, computing the hull of these vertices gives Convex-hull of entire polygon

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Question #3:

Method #1: Ray-Casting Algorithm (even-odd rule)

Algorithm:-

1- Choose a point p to test.
2- Draw a horizontal ray from p to right.
3- Count intersections b/w ray & polygon edges.
4- if no: of crossing is odd, point is inside.
5- if no: of crossing is even, point is outside.

Method #2:- Winding Number Algorithm.

Algorithm:-

1- Consider each consecutive pair of polygon vertices.
2- Calculate the signed angle through which direction from the test point to vertices changes.
3- Sum the signed angles around the polygon.
4- If total is approx $+2\pi$ or -2π then the point is inside.
5- If total is zero, the point is outside.

$$W(P) = \frac{1}{2\pi} \sum_{i=1}^{n} \theta_i$$

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Question #4:

Algorithm:

1. Take the input set of points.
2. Applying a clustering algorithm such as DBSCAN to divide the points into clusters say $C_1, C_2 \dots C_k$
3. For every cluster C_i, compute its Convex Hull H_i using Graham-Scan.
4. Compare the result hulls pair-wise using the polygon-intersection algorithm from Q1.
5. If no hull overlaps, return the hulls as multiple disjoint convex polygon.
6. If hulls overlap, report the overlap or revise the clustering parameters, depending on the requirements.

Pseudocode: Multiple-Convex-polygon(s):
    C = Cluster-Points(S)
    H = empty-list

    for each cluster C_i in C:
        H_i = Convex-Hull(C_i)
        add H_i to H
    for each pair H_i, H_j in H:
        if polygon-intersection(H_i, H_j)
            report Overlapping hulls
    return H;

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Question #1:

Pt (x,y)	(x,y)
P1	(13,0)
P2	(13,10)
P3	(10,10)
P4	(10,4)
P5	(2,10)
P6	(5,12)
P7	(2,5)
P8	(5,5)
P9	(2,7)
P10	(0,5)
P11	(2,4)
P12	(8,1)

Part(a): Brute force

Edge	Cross-product	Result
P10 → P14	37, 67, 55, 37, 43, 41, 28, 20 ...	all >0
P14 → P1	100, 97, 37, 54, 112, 44, 42 ...	all >0
P1 → P2	30, 30, 60, 80, 60, 80, 110	all >0
P2 → P6	80, 6, 54, 12, 52, 56, 46 x 66	all >0
P6 → P10	116, 66, 45, 75, 24, 49, 35	all >0

Rejected-pairs: P2 - P5, P10 - P12, P10 - P12

Hull: P10, P14, P1, P2, P6

Cost: 91 x 12 = 1092 sign tests => $O(n^3)$ Ans

part(b): Quick-Hull:

1- Extreme-points => A: P10 (0,5) B: P1 (13,0) A → B (13,-5)
Cross(P): 13(y - 5) + 5x

2- Value: P2 130, P3 115, P4 37, P5 100, P6 116, P7 35, P8 36, P11 -3, P12 -12, P13 -5, P14 -37 → Lower-set (negative)

3- Upper: farthest point is P2 (130)

4- Lower-point: the farthest point is P14 (37)

5- Hull: P10, P6, P2, P1, P14 => Avg $O(n \log n)$
Worst-Case $O(n^2)$

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part (iii): Divide & Conquer:

a) Sort by x, breaking ties by y: P10, P11, P9, P14, P8, P6, P13, P7, P5, P12, P4, P3, P1, P2.

Right-left-half (7) => Split into {P7, P5, P12} & {P4, P3, P1, P2}

Left-half (7) => Split into {P10, P11, P9} & {P14, P8, P6, P13}.

c) Merge by tangent:- Start with a = P13 (rightmost)
b = P7 (leftmost).

Upper tangent:
b moves P7 → P5 a-b (cross 5 > 0)
then a moves P13 - P8 (cross 18 > 0)
then b moves P5 → P2 (cross 12 > 0)
(P1: -80, P10: -66) → upper tangent is P6-P2.

Lower tangent:
b moves P7 → P12 (cross -7 < 0)
then a moves P13 → P14 (cross -5 < 0)
then b moves P12 → P1 (cross -5 < 0)
so lower tangent is P14-P1.

=> Hull: P10, P14, P1, P2, P6 $T(n) = 2T(n/2) + O(n)$
: $O(n \log n)$

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Part (iv): Graham-Scan

Points	Angle	Cost/Turn	Action	Stack after
P2	90°	-	push	P1, P2
P3	106.70°	P1, P2, P3 = 30	push	P1, P2, P3
P5	120.96°	P2, P3, P5 = 0	pop P3	P1, P2, P5
P6	123.69°	P2, P5, P6 = -12	pop P5	P1, P2, P6
P4	126.87°	P2, P6, P4 = 54	push	P1, P2, P6, P4
P7	146.15°	P6, P4, P7 = -19	pop P4	P1, P2, P6, P7
P9	147.53°	P6, P7, P9 = -31	pop P7	P1, P2, P6, P9
P8	148.00°	P6, P9, P8 = 21	push	P1, P2, P6, P9, P8
P10	158.96°	P9, P8, P10 = -10	pop P8	P1, P2, P6, P10
P11	160.62°	P6, P10, P11 = 15	push	P1, P2, P6, P10, P11
P13	164.05°	P10, 11, P13 = 0	push	P1, P2, P6, P10, P13
P12	168.65°	P10, P13, P12 = 0	pop P11	P1, P2, P6, P10, P12
P14	174.29°	P10, P12, P14 = -20	pop P13	P1, P2, P6 P10, P14

Hull: P1, P2, P6, P10, P14

Cost: O(nlogn) for the sort & O(n) for the scan

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Part (v):

Jarvis - march (gift - wrapping)

=) Start at the leftmost point p10.

- p10 → p14
- p14 → p1
- p1 → p2
- p2 → p6
- p6 → p10

Hull: p10, p14, p1, p2, p6 so h=5

Cost O(nh) = 14 × 6 = 70

Conclusion:

All five methods give the same hull

~~DFF~~ p10 (0,5), p14 (3,1), p1 (12,6), p2 (13,10), p6 (8,12)

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Q#6:

Weather stations are:

A(-1,0) B(1,3) C(3,3) D(5,4) E(7,1) F(5,-3)
G(2,-4) H(-2,-2) I(2,1) J(3,0) K(1,1) L(4,2)

Left-most-point is H(-2,-2)
Right-most-point is E(7,1).
Base-orient (H E D)
= 9(y+2) - 3(x+2) = 9y - 3x + 12

Point	orient (H, E, D)	side
A (-1,0)	15	+
B (1,3)	36	+
C (3,3)	48	+
D (5,4)	33	+
E (7,1)	0	line endpoint
F (5,-3)	-30	-
G (2,-4)	-30	-
H (-2,-2)	0	line endpoint
I (2,1)	15	+
J (3,0)	3	+
K (1,1)	18	+
L (4,2)	18	+

=) C has largest +ve cross-product (48). while F & G lie on the -ve side. Final hulls H → G → F → E → D → C → B → A, Interior-points I, J, K, L are excluded. Verification: 97/2 = 48.5 sq-units

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Question #7.

a) Segments: $S1$: (1,1) - (5,5)
$S2$: (1,5) - (5,1)
$S3$: (5,5) - (5,1)
$S4$: (2,2) - (4,4)
$S5$: (0,0) - (0,4)

Part(i)

Pair	Result	classification/point
S1, S2	Intersect	proper-crossing at (3,3).
S1, S3	Intersect	Endpoint touch at (5,5)
S1, S4	Overlap	S4 lie entirely on S1
S1, S5	Disjoint	No - intersection.
S2, S3	Intersect	Endpoint touch at (5,1)
S2, S4	Intersect	proper crossing at (3,3)
S2, S5	Disjoint	No - intersection.
S3, S4	Disjoint	No - intersection.
S3, S5	Disjoint	No - intersection
S4, S5	Disjoint	No - intersection.

Part(ii):

Algorithm (Brute-force approach)

1- Consider every pair of segments (S_i, S_j); $i \neq j$
2- Apply a robust segment intersection test using orientation tests
3- if they intersect, report the pair & intersection type.

Time-Complexity: $\frac{n(n-1)}{2} \Rightarrow O(n^2)$

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part(iii):-

for n = 100,000 line segments

$\frac{n(n-1)}{2} = 4999950000 \approx 5 \times 10^9$ (too-expensive).

2) Use - Bentley Ottmann Sweep-line algorithm which reports intersection in typically $O((n+k)\log n)$ time where k is the number of intersections.

Limitation:- Reporting k intersections requires at least $O(k)$ time.

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Question #8:-

n = 100,000 brute force vs Graham vs Jarvis

$\log_2(100000) = 16.61$

Algorithm	Expression	Count for $n = 10^5$
Brute-force	n^3	10^{15}
Graham-scan	$n \log_2 n = 10^5 \times 16.61$	1.66×10^6
Jarvis	$n \cdot h = 10^5 \times 100$	10^7

a) Which Algo to choose:-

- Simple Computation: Use Graham-scan or Andrews Monotone Chain, both $O(n \log n)$. Jarvis March is preferable when the no: of hull vertices h is very small $O(nh)$.

- Repeated insertions: Use an incremental/dynamic convex hull. Test whether a point is inside in $O(\log h)$; otherwise, find tangents & update the hull. Rebuilding from scratch costs $O(n \log n)$ per insertion.

c) Implementation pitfalls:-

1- Numerical Overflows: Cross products can overflow 64-bit int.
2- Duplicate - Overflows Remove duplicates to avoid incorrect results.
3- Collinear-points: Use Consistent cross-product condition to keep or discard collinear points.
4- Degenerate - Cases: Handle all - collinear points separately.

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Question #9:-

2) Checking boundary & then using ray-casting.

(i) P(4,4): P is not on any edge.

2) Using the standard vertex-counting convention. There is one intersection to right

1 is odd so P is inside.

(ii) Q(8,6): lies exactly on edge CD

→ the line through C(9,4) & D(7,8) has equation

$y = 2x + 22$ $x=8$ it gives $y=6$

Since Q is on segment CD, Q is on the "Boundary".

(iii) R(16,5):

R lies to the right of the polygon's rightmost vertex C(9,4). The rightward ray crosses no polygon edge. so the parity is even & R is outside.

(iv) S(4,1):

AB has $y=1$. S(4,1) satisfies $y=1$ & $1 \leq 4 \leq 7$ so S lies on edge AB & it on the "boundary."

P (inside)
Q (Boundary)
R (outside)
S (Boundary)

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Calculations:

- S(4,1): edge A→B has direction (6,0)
  S-A = (2,0)
  cross = 0
  1 ≤ 4 ≤ 7 on the boundary.

- Q(8,6): edge C→D has direction (-2,4)
  Q-C = (-1,2)
  cross = (-2)(2) - 4(-1) = 0
  Q is on the boundary (edge CD)
  CD is y = 4 + 2(9-x)
  at x = 8, y = 6. Ans.

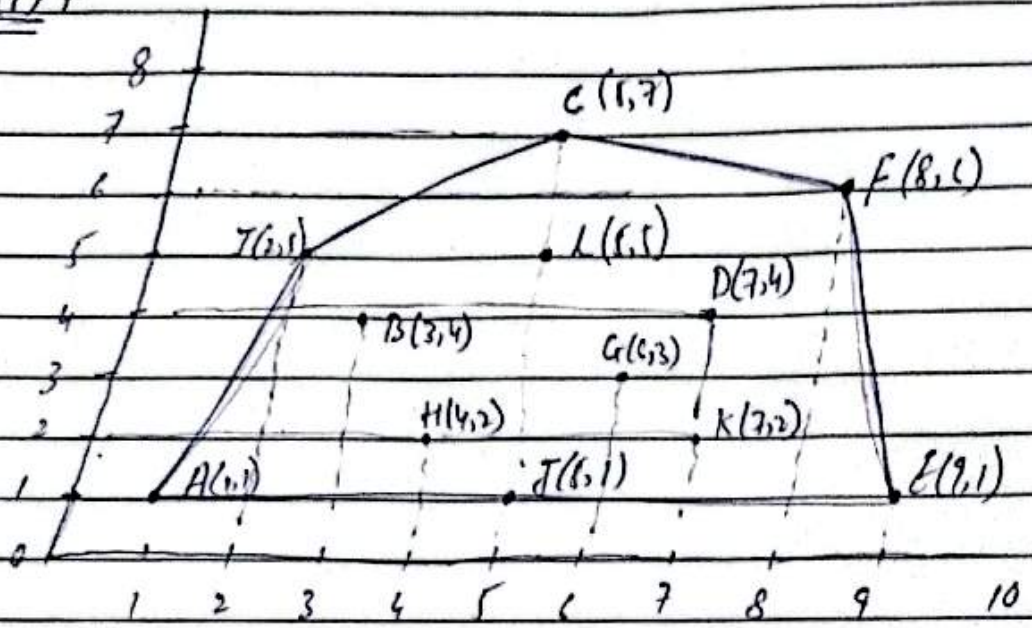
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Question #10:

part(i):



Ans.

part(ii):-

Extreme points:-

- Left-most: A(1,1)
- Right most: E(9,1)
- Lowest: y=1 is shared by A, J & E
- Highest: C(5,7)

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part(iii) & part(iv):

Brute-force with the Orientation-formula:

Orient(P,Q,R): $(Qx-Px)(Ry-Py) - (Qy-Py)(Rx-Px)$

Edge	formula	Values	Result:
A→E	8(y-1)	B24, C48, D24, F40, G16, H8 T32, J0, K8, L32	all >0
E→F	46-y-5x	A46, B27, C14, D7, G13, H24, I31, J20, K9, L16	all >0.
F→C	26-3y-x	A22, B11, D7, E14, G11, H16, I9, J18, K13, 26	all >0
C→I	11+2x-2y	A10, B5, D13, E26, F9, G14, H13, J18, K19, L6	all >0.
I→A	4x-y-3	B5, C10, D21, E32, F23, G18, H11, J16, K23, L12	all >0.

Rejected-pairs: A-C(4,6). D gives -24 & I gives +16.
B-D(4,0), C gives +12 A gives -12

Collinear-handling: J(5,1) = 0
A-J J-E
A-E J is a boundary-point.

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Hull(ccw): $A(1,1) \to E(9,1) \to F(8,6) \to C(5,7) \to I(2,5) \to A$

=) Turn check:-

Turn	Cross
$A \to E(8,0)$ with $E \to F(-1,5)$	+40
$B \to F$ with $F \to C(-3,1)$	+14
$F \to C$ with $C \to I(-3,-2)$	+9
$C \to I$ with $I \to A(-3,-2)$	+16
$I \to A$ with $A \to E$	+32.

part(v): pivot is $A(1,1)$ → lowest (leftmost).

Graham-Scan

Point	Vector from A	Angle
J	(4,0)	0°
E	(8,0)	0°
K	(6,1)	9.46°
H	(3,1)	18.43°
G	(5,2)	21.80°
D	(6,3)	26.57°
F	(7,5)	35.54°
L	(4,4)	45°
B	(2,3)	56.31°
C	(4,6)	56.31°
I	(1,4)	75.96°

=) J & B are dropped.

Next	Cross	Action	stack
E	-	push	A, E
K	A,E,K=8	push	A, E, K.
H	E,K,H=13	push	A, E, K, H.
G	K,H,G=-3	pop H	A, E, K, G.
D	K,G,D=-5	pop G	A E D
F	E,D,F	pop D	A, E, F
L	E,F,L	push	A, E, F, L
C	F,L,C	pop L	A, E, F, C
I	F,C,I	push	A, E, F, C, I

final hulls $A \to E \to F \to C \to I$

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part(vi). Comparison with Quick-hull:-

Quick-Hull:-

- Initial-line: Take A(0,1) & E(9,1)
  $\delta(y-1)$ → cross-product
  Points above the line are on the upper-side.
  J lies on the line.

- farthest-point: C has max cross-product = 48
- line A→C: I = 16 is outside so choose I
  Edges A→I &
  I→C have no points outside & are finalized.
- line C→E: F = 44 is outside.

Final-Hull:- A→I→C→F→E (plus J on edge AE).

Comparison:-

Methods	Hull-Vertices
Brute-force	A E F C I
Graham-Scan	A E F C I
Quick-hull.	A E F C I

∴ All three agree Ans.

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Part (vii):-

Algorithms	Complexity
Brute-force	$O(n^3)$
Graham-scan	$O(n \log n)$
Quick-Hull	$O(n \log n)$ avg. $O(n^2)$ Worst-Case.

Part (viii):-

Edge	$x_i \cdot y_{i+1} - y_i \cdot x_{i+1}$	Value
$A \to E$	$1.1 - 1.9$	-8
$E \to F$	$9.6 - 1.8$	46
$F \to C$	$8.7 - 6.5$	26
$C \to I$	$5.5 - 7.2$	11
$I \to A$	$2.1 - 5.1$	-3.

Sum = 72.80, Area = $\frac{1}{2} 72 = 36$ sq. unit

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